

FROM HAND TO HAND: A Shared Wearable Interface and Functional Benchmark for Dexterous Manipulation

Author names omitted for anonymous review

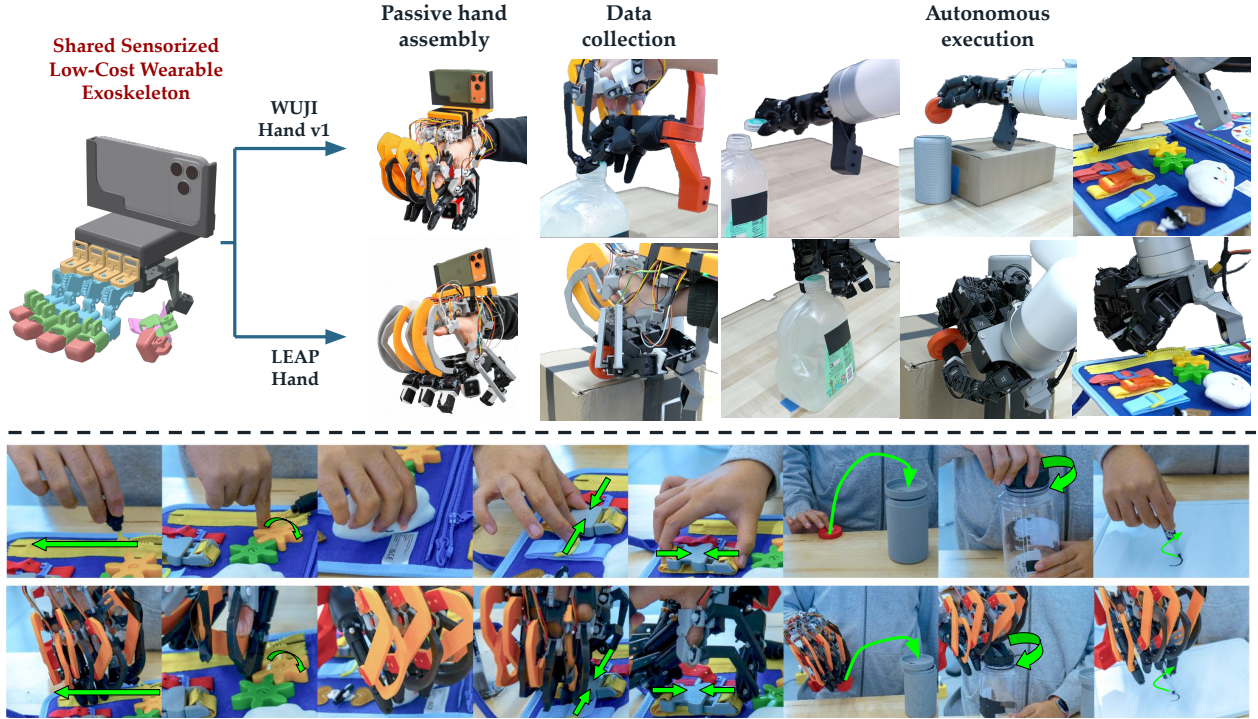


Fig. 1: **Top:** Overview of WUDI. A shared sensorized exoskeleton supports different robot hands through interchangeable passive hands and linkages, enabling demonstration collection and downstream policy learning. **Bottom:** The eight OTHER benchmark tasks, performed bare-handed (upper row) and through WUDI (lower row). From left to right: Zipper, Spin, Squeeze, Unbuckle, Buckle, Coin, Lid, and Write. Green arrows indicate task motions.

Abstract—Collecting demonstrations for dexterous manipulation is difficult to scale across robot hands with different kinematics and contact geometry. Existing interfaces often require embodiment-specific sensing hardware, motion retargeting, or visual correction. We present WUDI, a Wearable, Unified Interface for Dexterous manipulation that separates a shared sensorized exoskeleton from interchangeable passive hands and linkages. This design reuses the sensing hardware across embodiments and supports robot-free demonstration collection without visual correction within the supported workspace. We evaluate demonstration collection and policy learning with WUJI Hand v1 and LEAP Hand. To assess functional manipulation, we introduce OTHER, an eight-task suite inspired by occupational therapy that spans precision grasping, tool use, pulling, rotation, compression, and fastening. A 12-participant study evaluates usability and task performance, and demonstrations collected with WUDI support autonomous policies on three tasks across both hands. In an expert-operator comparison on LEAP Hand, WUDI yields 3.62–7.71× more successful task completions per collection window than a vision-based teleoperation baseline. These results demonstrate a practical approach to reusing a wearable sensing interface across robot hands while exposing the effects of mechanical constraints on functional dexterity. Project Website: <https://wudiwudiwudi2026.github.io/>

I. INTRODUCTION

Everyday manipulation involves more than picking and placing objects: opening a bottle, pulling a zipper, or inserting a buckle requires coordinated finger motion, sustained contact, and controlled force. Achieving this class of behavior with robot hands remains an open problem, and two practical bottlenecks hold back progress. The first is data: learning these behaviors from real-world demonstration requires large amounts of expert data suited to the target robot hand. The second is evaluation: without a common task suite, it is difficult to say whether a new hand, interface, or policy represents progress over prior work. Both bottlenecks share a root cause: robot hands differ in joint structure, link geometry, and available motion, so demonstrations do not transfer between embodiments and evaluation results are not comparable across them.

On the data side, existing interfaces establish correspondence mechanically [1]–[3]. Because sensing and embodiment-specific structure are built into the same device, supporting a new hand can require additional mechanical

redesign. Evaluation faces a parallel problem. Dexterous hands are typically demonstrated on task sets chosen alongside the hardware, so results are not comparable across systems. Because human-operated performance is rarely reported alongside autonomous performance, hardware limits are not separated from learning limits, and it is unclear what a given result says about the field’s progress.

Physical task boards are the established answer to comparability in manipulation benchmarking. NIST targets industrial assembly [4], while Robothon and euROBIN use instrumented boards to compare completion time and success across laboratories and between humans and robots [5]. These benchmarks primarily assess arm-level skills with parallel-jaw grippers, rather than multi-finger coordination across hand embodiments. We also take seriously the reproducibility critique leveled at boards assembled from many distinct parts, which are difficult to rebuild and therefore limit who can run the benchmark [6].

Our key insight has two parts. First, a generalizable data-collection interface should separate sensing hardware from embodiment-specific geometry, and should provide sufficient degrees of freedom (DoF) to accommodate different hands. Without both, adapting an interface to a new hand can require redesigning the sensorized components and exoskeleton, rather than only the passive parts. Second, the problem of comparing functionally different hands on shared tasks is not new. Occupational therapy (OT) has developed validated instruments for exactly this. The Sollerman Hand Function Test [7] scores twenty activities of daily living: what makes it portable is that it is grounded in functional outcome rather than in a particular anatomy. We argue that the same property makes it a natural basis for measuring robot dexterity. Supporting motion is not the same as being dexterous, and the available motion must be shown to enable functional manipulation.

We present WUDI, a wearable interface that implements this separation through a sensorized exoskeleton, interchangeable passive hands, and embodiment-specific linkages. Adapting the interface to a new hand requires selecting coupler dimensions and checking mechanical compatibility, and retains the sensing hardware unchanged. Fixed or coupled joints can reduce independent degrees of freedom and reachable workspace. To assess functional dexterity across embodiments, we introduce the Occupational Therapy-inspired Hand Evaluation and tRaining (**OTHER**) benchmark. OTHER combines activities adapted from the Sollerman Hand Function Test with busy-board tasks that consolidate the fastening, latching, and rotation activities of OT practice into a single reproducible fixture. It evaluates both human-operated demonstration collection and autonomous execution, so that the performance reachable by a human teleoperator on a given hand serves as a reference for what a policy on that hand could achieve.

Our contributions are:

- A wearable demonstration interface that shares a single sensorized exoskeleton across robot hands by confining embodiment-specific correspondence to interchangeable

passive components.

- A synthesis pipeline and mechanism-based joint reconstruction that produce demonstrations directly in the target hand’s joint coordinates.
- OTHER, an eight-task benchmark adapted from occupational therapy hand function assessment, with evaluations of human-operated manipulation and downstream policy learning across two robot hands.

II. RELATED WORK

A. Mechanically Coupled Demonstration Interfaces

Mechanically coupled interfaces establish correspondence with the target robot in hardware. UMI [8] enables portable, robot-free collection for gripper manipulation, while Dex-UMI [3] extends this idea to dexterous hands using a wearable exoskeleton and visual correction. DEXOP [1] and DexEXO [2] use passive replica hands to align operator motion with robot kinematics and contact geometry, and RealDexUMI [9] uses an isomorphic glove for direct joint-space control. These approaches reduce software-retargeting error but each is presented for particular hands (Inspire Hand, XHand, or OYMotion ROHand). WUDI instead shares a single sensorized exoskeleton across embodiments and confines adaptation to interchangeable passive hands and linkages (Table I).

B. Human-Motion Capture and Retargeting

Human-motion interfaces capture demonstrations and re-target them to robot hands. AnyTeleop and TeleVision [10], [11] use vision, DOGlove and ART-Glove [12], [13] use instrumented gloves, and DexCap [14] supports robot-free motion capture. Transfer must accommodate differences in morphology, joint limits, and feasible contacts. Existing methods address these differences through trajectory retargeting [15], object-guided reinforcement learning [16], embodiment-agnostic intent learning [17], residual reinforcement learning [18], or physics-informed retargeting [19]. These methods introduce an optimization or learning stage to map human motion into robot actions, whereas WUDI establishes correspondence mechanically within its supported workspace, making the reconstruction of passive hand joints simpler (Sec. III-C). Table I compares the interfaces.

C. Dexterous Manipulation Benchmarks

Dexterous manipulation benchmarks span physical task boards, simulation, and hardware evaluations. The NIST Assembly Task Boards [4] and Robothon/euROBIN [5] support standardized comparison of assembly skills, but primarily target arm-level manipulation with parallel-jaw grippers. Simulation benchmarks enable controlled evaluation across tasks and embodiments [22]–[26], while physical performance is additionally affected by backlash, actuation uncertainty, and contact compliance. POMDAR [27] evaluates 18 physical tasks across ORCA hand embodiments, but relies on teleoperation rather than learned policies. Outside robotics, the Sollerman Hand Function Test [7] evaluates everyday activities such as zippers, buttons, and jars. Inspired by these

TABLE I: Comparison of data collection interfaces.

Method	No Vis. Adapt.	Diff. Hand	Port.	No Retarg.	Full ROM
<i>Mechanically Coupled / Isomorphic</i>					
DexUMI [3]	×	×	✓	✓	×
RealDexUMI [9]	✓	×	✓	✓	×
DEXOP [1]	✓	×	✓	✓	×
DexEXO [2]	✓	×	✓	✓	×
<i>Human-Motion Capture / Glove-Based</i>					
DexCap [14]	×	○	✓	×	✓
AnyTeleop [10]	✓	○	×	×	✓
TeleVision [11]	✓	✓	×	×	✓
DOGlove [12]	✓	○	×	×	×
OSMO [20]	✓	○	✓	×	✓
ACE [21]	✓	✓	×	×	✓
WUDI	✓	✓	✓	○	×

No Vis. Adapt.: no embodiment-gap correction of visual observations, such as inpainting or hand-geometry replacement. *Diff. Hand*: reuse of the sensing interface across different robot hands; replacing embodiment-specific passive parts is allowed. *Port.*: portable collection without a robot arm. *No Retarg.*: no human-to-robot hand-motion retargeting, excluding calibration of joint encoder. *Full ROM*: Hand motion is not mechanically constrained. ✓ = yes, ○ = limited or not established, × = no.

functional assessments, OTHER combines eight everyday manipulation tasks into a shared physical benchmark for human-operated demonstration collection and downstream policy learning across robot hands.

III. THE WUDI INTERFACE

WUDI couples a shared sensorized exoskeleton to an interchangeable passive hand through embodiment-specific linkages (Fig. 2). The passive hand reproduces the target’s supported articulation and contact geometry; all encoders and wiring remain on the exoskeleton. This separation enables hardware reuse but requires inferring passive-hand configurations through the linkage mechanism. We describe the design requirements, mechanical synthesis, wrist sensing, and motion-reconstruction software below.

A. Design Requirements

R1 (Kinematic correspondence). Preserve target joint axes and link geometry within the supported articulation, applying identical fixed-angle and coupling constraints during collection and robot execution.

R2 (Shared sensing). Retain the exoskeleton, encoders, and electronics across hands; adapt only passive components and their synthesis/reconstruction parameters, with compatibility checks and manual adjustments where needed.

R3 (No embodiment-specific visual correction). Use target-shaped passive hands and matched wrist viewpoints to support policy learning without segmentation, inpainting, or rendered hand replacement.

R4 (Portable and Scalable operation). Support robot-free collection after brief familiarization, limiting per-hand hardware additions to printed components and fasteners.

B. Mechanical Interface and Sensor Synthesis

1) *Shared exoskeleton*: To satisfy R2, all encoders and electronics remain on the shared exoskeleton when the

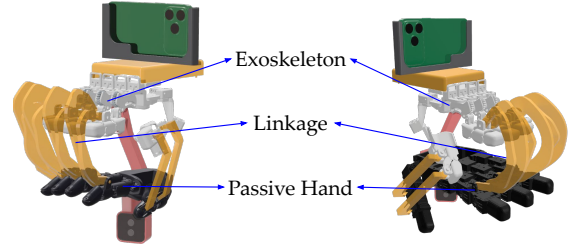


Fig. 2: The three components of WUDI, shown for WUJI Hand v1 (left) and LEAP Hand (right): a shared sensorized exoskeleton connects to interchangeable passive hands through embodiment-specific linkages.

passive hand is replaced. For each of four non-thumb fingers, it accommodates metacarpophalangeal (MCP) abduction/adduction (AA) and flexion/extension (FE), together with proximal interphalangeal (PIP; middle finger joint) flexion. Three additional thumb motions provide 15 operator-side degrees of freedom in total. The thumb mechanism accommodates carpometacarpal (CMC; thumb-base), MCP, and interphalangeal (IP) rotation. Sixteen resistive rotary sensors measure exoskeleton joint angles or linkage rotation relative to the supporting segment (Fig. 3(a)). Because these measurements are not generally identical to passive-hand joint angles, we recover the latter using the mechanism model in Sec. III-C.

2) *Interchangeable passive hand*: To satisfy R1, we derive passive-hand geometry from the target’s kinematic description and link meshes. The automated pipeline normalizes link poses, preserves joint-axis metadata, and exports a named, exact-faceted STEP assembly through FreeCAD, validating solid integrity after export. Joint-core installation, linkage-mount construction, interference relief, and mechanical stops are completed through reviewed CAD edits. Manual thumb adaptation (Sec. III-B.4) completes the assembly. To satisfy R2, all sensors and electronics remain on the shared exoskeleton.

3) *Non-thumb linkage construction*: To satisfy R1 and R2, embodiment-specific linkages couple the passive hand to the shared sensorized exoskeleton, accommodating MCP-FE, MCP-AA, and PIP flexion. Following DEXOP [1], two serial four-bar linkages transmit MCP/PIP flexion. Standoffs fix the base separation; the first coupler connects the two PIP axes and serves as the second linkage’s moving ground. The second coupler connects middle-link mounts with matching offsets in their respective PIP frames (Fig. 3(b)–(c)).

(a) **Relative placement and coupling geometry**: Each embodiment uses a fixed CAD-defined placement between the exoskeleton and passive-hand bases, described by the fore-aft offset x and vertical separation h . This placement determines the reference attachment geometry and pin-to-pin lengths of the coupling bars.

(b) **Finger layouts**. The passive hand retains its embodiment-specific finger orientations. For tilted layouts, such as WUJI, coupling to the parallel exoskeleton can induce additional AA motion during flexion. Reconstruction

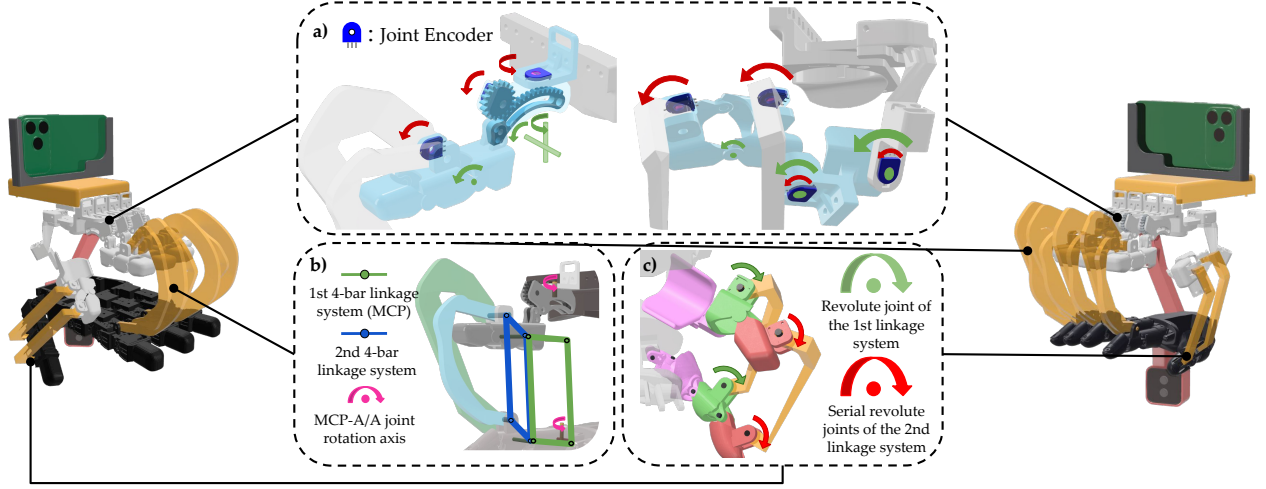


Fig. 3: Mechanical design and sensing of WUDi, shown with LEAP (left) and WUJI (right). (a) Exoskeleton finger and thumb mechanisms, with joint encoders highlighted in blue. (b) Two serial four-bar linkages transmit MCP and PIP flexion. (c) Revolute joints connecting the linkage components.

therefore uses the spatial mechanism geometry rather than assuming identical exoskeleton and passive-hand abduction angles. For target hands with a distal interphalangeal (DIP) joint, we fix it at 30° in both the passive replica and the deployed robot.

4) *Thumb adaptation*: Thumb metacarpal layouts and axes vary substantially across hands, so the thumb linkage is designed per embodiment to satisfy R2. The exoskeleton provides CMC, MCP, and IP rotation; two couplers connect mount positions, orientations, and coupler lengths to accommodate this variation while preserving useful grasping motion.

5) *Wrist-mounted sensing*: To support R3, a wrist-mounted camera provides corresponding viewpoints during demonstration collection and robot execution. An iPhone streams six-degree-of-freedom ARKit poses over USB via Record3D, while a RealSense D405 records RGB-D observations. A keyed mount maintains the phone-to-palm extrinsic transform. Fig. 4 compares wrist-camera observations during passive-hand demonstration collection and execution with the corresponding robot hand. Together, these onboard measurements support the robot-free collection required by R4.

C. Software Interface

The software converts exoskeleton sensor readings into target-hand joint trajectories through encoder calibration and mechanism-based joint reconstruction, then aligns them with visual and wrist-pose observations for policy learning. Raw measurements are retained to support reprocessing without repeating data collection.

1) *System reconstruction*: The exoskeleton, passive hand, and linkages are represented as a unified kinematic model in MuJoCo. Registering the passive hand to the source robot model defines a common reference configuration and preserves the target joint axes and ordering. For joint j , let $\mathbf{c}_j \in \mathbb{R}^3$ be a point on its rotation axis, $\mathbf{a}_j \in \mathbb{R}^3$ the unit



Fig. 4: Wrist-camera observations during demonstration collection with the passive hand and deployment with the corresponding robot hand.

axis direction, and Δq_j its displacement from the reference configuration. Its rigid transform is

$$G_j(\Delta q_j) = \begin{bmatrix} R_j & (I - R_j)\mathbf{c}_j \\ \mathbf{0}^\top & 1 \end{bmatrix},$$

$$T_{WL}(\Delta \mathbf{q}) = \left(\prod_{j \in \mathcal{A}(L)} G_j(\Delta q_j) \right) T_{WL}^0, \quad (1)$$

where $R_j = \text{Rot}(\mathbf{a}_j, \Delta q_j)$ is the rotation about axis $\mathbf{a}_j$, T_{WL}^0 is the reference pose of link L in the world frame W , and $\mathcal{A}(L)$ is the ordered set of joints from the root to

L. The same forward-kinematic model propagates the poses of the hand links, linkage bars, and their attachment points, enabling mechanism constraints to be evaluated during state reconstruction.

2) *Encoder calibration*: Per-channel affine mappings convert ADC counts into joint or carrier-relative coupler angles, accounting for direction, offset, and transmission gain. Geared MCP sensors use two mechanical stops 50° apart. Sensor calibration is reused across passive hands, while a reference-pose capture registers each assembly.

3) *Mechanism-based joint reconstruction*: Direct-joint measurements fix measured exoskeleton coordinates. Bounded nonlinear least squares reconstruct the remaining coordinates, passive-hand angles, and rigid-bar poses from mount-angle errors and radial, axial, and angular pin residuals, allowing finite assembly clearance. Fixed joints retain their prescribed angles. Warm starts and continuity regularization stabilize reconstruction; excessive residuals or joint increments trigger rejection. Residuals and Jacobian diagnostics record constraint consistency and local ambiguity.

4) *Data acquisition and processing*: Encoder, phone, and RGB streams are recorded independently on a shared clock. Calibrated encoder signals are smoothed with a Hampel filter followed by a One Euro filter [28] before reconstruction. Reconstructed displacements are converted into absolute hand targets using reference angles and axis conventions. Camera and phone samples are matched to trajectory timestamps by nearest-neighbor, retaining temporal offsets and validity flags for downstream policy training.

D. Hardware cost

To satisfy R4, WUDI uses a reusable sensing assembly and printed passive components. The complete system, including 16 rotary encoders, acquisition electronics (microcontroller and ADC), custom PCB, and one spool of PLA-CF filament sufficient for the exoskeleton and one passive-hand assembly, costs approximately \$120. Each additional hand adds only printed parts and fasteners. The wrist-mounted RealSense D405 and the iPhone for pose tracking are excluded.

IV. THE OTHER BENCHMARK

We introduce the Occupational Therapy-inspired Hand Evaluation and tRaining benchmark (OTHER), an eight-task suite that evaluates functional dexterity during human-operated demonstration collection and autonomous robot execution. The tasks require coordinated finger motion and sustained object contact across precision grasping, tool use, pulling, rotation, compression, and fastening. Coin, Lid, and Write are adapted from activities in the Sollerman Hand Function Test [7], while Zipper, Spin, Squeeze, Buckle, and Unbuckle use common busy-board mechanisms. Table II summarizes the eight tasks and their completion conditions. Each condition is defined by the object-level outcome, allowing hands with different kinematics to use different manipulation strategies. Fig. 1 illustrates the tasks performed bare-handed and through WUDI. We evaluate

human-operated performance on all eight tasks and autonomous policies on Zipper, Lid, and Coin, selected to represent the easy, medium, and hard groups observed in our user study. Demonstrations for the challenging group could not be collected because operators rarely succeeded. Human-operated results characterize the capabilities and limitations of the interface-hand combination, providing context for interpreting downstream policy performance. We will open source the real world and simulated benchmark assets on our website.

TABLE II: The eight OTHER tasks' completion conditions and principal dexterity demands

Task	Completion condition	Dexterity demand
Coin	Pick up a coin and deposit it in the jar.	Precision grasp and release.
Lid	Fully unscrew the bottle lid.	Grip-torque coordination.
Write	Write an "S" with the marker.	Tool grasp and contact control.
Zipper	Pull the slider to the end of the track.	Grasp and directionally constrained pulling.
Spin	Rotate the wheel by 180° .	Rotational control.
Squeeze	Compress until it sounds.	Forceful compression.
Buckle	Insert and latch the buckle.	Alignment and insertion.
Unbuckle	Release the buckle latch.	Coordinated latch actuation.

V. EXPERIMENTS

Prior mechanically coupled interfaces tie their sensing hardware to a single robot hand. WUDI aims to remove this dependency through a reusable sensing interface, and our experiments address one central question: *Can the same exoskeleton and sensing hardware support effective demonstration collection across different robot-hand geometries?* Limited access to the robot hands required by prior embodiment-specific exoskeletons [1]–[3] restricts our direct comparisons with these systems. We therefore evaluate: **Q1: Usability** Can users perform everyday tasks comfortably and effectively? **Q2: Policy learning** Do collected demonstrations enable autonomous manipulation? **Q3: Reconstruction accuracy** How accurately are passive-hand joint angles recovered? **Q4: Workspace preservation** How much fingertip workspace remains accessible? **Q5: Collection efficiency** How does throughput compare with vision-based teleoperation and bare-handed operation?

A. User Study

We evaluated the WUDI Hand v1 configuration of WUDI with 12 participants (6 male and 6 female) with an IRB-approved procedure. Hand length ranged from 15.5 to 19.9 cm, and hand width ranged from 8.0 to 11.0 cm. Participants attempted each of the eight OTHER tasks once bare-handed for familiarization. We measured donning time, followed by five minutes of free practice. Participants attempted five consecutive trials per task in a randomized task order, yielding 40 planned trials per participant. Trials ended upon success, timeout, participant request, or a state requiring reset. After a timeout, the attempt was recorded as unsuccessful, and participants could choose to retry the same task or move on to the next. Participants then completed a questionnaire on 5-point Likert scale and provided qualitative

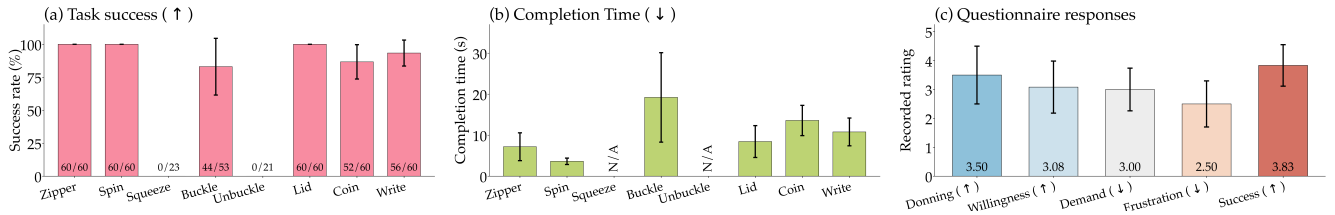


Fig. 5: User-study results with WUDI ($n = 12$). **(a)** Task success rates over recorded attempts ($\uparrow$, higher is better) **(b)** Completion time on successful trials ($\downarrow$, lower is better): for each participant and task, successful trial times were averaged, then summarized across participants as mean $\pm$ sample standard deviation; N/A denotes no successful attempts. **(c)** Questionnaire ratings (mean $\pm$ sample standard deviation across 12 respondents); arrows indicate the preferred direction for each item. Demand and frustration are shown in their original direction and are not reverse-coded.

feedback. We quantify uncertainty using participant-level bootstrap resampling, retaining all trials from each sampled participant to preserve within-participant dependence. Completion-time estimates exclude unsuccessful trials.

Fig. 5 (a–b) summarizes 397 recorded attempts; 83 unrecorded slots were excluded from success-rate denominators. Success rates were 100% on Zipper (60/60), Spin (60/60), and Lid (60/60), 83.0% on Buckle (44/53), 86.7% on Coin (52/60), and 93.3% on Write (56/60), with no successes on Squeeze (0/23) or Unbuckle (0/21). Buckle had the longest mean successful completion time (19.21 s; 95% CI: 13.13–26.00 s). Under the optional-continuation protocol, these results motivate the easy (Zipper, Spin), medium (Lid, Write), hard (Buckle, Coin), and challenging (Squeeze, Unbuckle) groups for the evaluated interface.

Donning and doffing took 16.81 ± 6.27 s and 4.05 ± 3.98 s, respectively (mean $\pm$ SD). Across 12 respondents, perceived success averaged $3.83/5$ and willingness to use the interface $3.08/5$ (Fig. 5(c)). Brief setup and successful manipulation with mechanical feedback support practical robot-free demonstration collection, while willingness ratings indicate room to improve the user experience.

B. Policy Learning with WUDI

1) *Embodiment and Task Setup*: We evaluate WUDI on two robot embodiments and three dexterous manipulation tasks. We build WUDI data-collection interfaces for the WUJI Hand v1 and LEAP Hand. For policy evaluation, we mount each robot hand on a UFactory xArm 6. We select Zipper, Lid, and Coin as one task from each of the easy, medium, and hard groups observed in the user study.

2) *Demonstration Collection and Policy Training*: We collect 70 demonstrations using WUDI per task–hand pair and train a separate policy for each. We reconstruct the passive hand’s joint configuration from these measurements to use as action space. We train a standard Diffusion Policy [29] with observation consists of wrist camera RGB stream together with relative robot end-effector (EE) pose. The action consists of relative arm EE action and absolute hand poses. We use a ResNet-18 visual encoder and the standard temporal U-Net Diffusion Policy backbone.

3) *Results and Analysis*: The policy evaluation and success criteria follow Sec. IV. As shown in Table III, WUJI achieves 80% success on Zipper, 75% on Lid and 45% on

Coin, while LEAP achieves 55% on Zipper, 65% on Coin and only 15% on Lid. LEAP’s bulkier geometry facilitates pinch grasps, consistent with its higher Coin success. In contrast, WUJI’s smaller geometry can prevent full finger closure for operators with larger hands. These results demonstrate policy learning across both embodiments while highlighting task-dependent effects of hand geometry and operator fit. LEAP’s low success on Lid is notable. It is not a capability limit: an expert completes Lid 11 times per window with the same configuration (Sec. V-E), and reconstruction error (Sec. V-C) and thumb workspace (Sec. V-D) are sufficient for the grasp. Instead, 16 of the 17 failures closed the fingers past contact and applied excessive force, ejecting the lid. WUDI records positions but not forces, so the policy cannot modulate grip after contact.

TABLE III: Policy success rates across robot hands and tasks.

Hand	Zipper	Lid	Coin
WUJI Hand v1	16/20(80%)	15/20(75%)	9/20(45%)
LEAP Hand	11/20(55%)	3/20(15%)	13/20(65%)

C. LEAP Joint Reconstruction Accuracy

We quantitatively evaluate WUDI’s mechanism-based joint reconstruction on the LEAP passive hand using motion-capture ground truth. This experiment rules out reconstruction error as a cause of the low LEAP–Lid success (Sec. V-B). We evaluate the thumb and one representative non-thumb finger (index), since the three LEAP non-thumb fingers share the same kinematic and linkage structure. Motion-capture markers are attached to the passive hand to recover ground-truth joint angles. We compare these measurements with the joint configurations reconstructed by WUDI and report mean absolute error (MAE), root-mean-square error (RMSE), and maximum absolute error.

Table IV reports MAEs of 1.34° – 4.20° , RMSEs at most 5.50° , and maximum errors of 9.27° – 14.06° . MCP-FE has the largest average error. These results support accurate reconstruction of the evaluated LEAP joints over the tested motion range.

D. Fingertip Workspace Coverage

We compare reconstructed fingertip regions from WUDI and vision-based teleoperation with each hand’s

TABLE IV: Joint reconstruction error

Finger	Joint	MAE (°)	RMSE (°)	Max (°)
Thumb	MCP	3.17	3.97	10.09
	IP	3.33	4.55	12.84
Index	MCP-AA	2.10	2.89	10.23
	MCP-FE	4.20	5.50	14.06
	PIP	1.34	2.33	9.27

native workspace. Regions are reconstructed using Delaunay tetrahedra with circumsphere radii up to 25 mm and rasterized at 2 mm resolution. We report the ratio of reconstructed fingertip volume under WUDI to that obtained from sampled vision-based teleoperation commands. The latter represents robot configurations produced by human-motion retargeting within the robot’s joint limits, without WUDI’s mechanical constraints. Fig. 6 shows that WUDI covers a comparable reconstructed fingertip volume to the vision-based baseline, but only 0.27 for the LEAP thumb. This reflects the LEAP configuration, which fixes non-thumb DIP joints at 30° and locks both thumb-root joints, whereas the vision baseline does not impose these constraints. Vision-based teleoperation achieves greater coverage in all four comparisons. Section V-E evaluates whether this reduced workspace nevertheless supports efficient task completion.

	LEAP		WUJI	
	Thumb	Non-thumb	Thumb	Non-thumb
WUDI / Vision	0.27	0.84	0.83	0.58

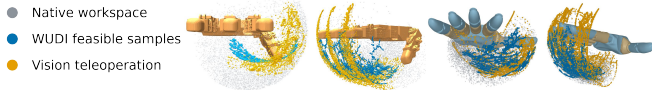


Fig. 6: Fingertip workspaces for LEAP and WUJI under native articulation, WUDI constraints, and vision-based teleoperation.

E. Efficiency Analysis

We compare WUDI with a vision-based data collection baseline on Zipper, Coin, and Lid. This baseline estimates the 3D hand pose with HaMeR [30] from multi-view RGB streams, and retargets the hand pose to robot joints to teleoperate the robot in real time. Both interfaces use LEAP, chosen because its compliance tolerates the pose errors inherent to vision-based retargeting, giving the baseline its most favorable conditions. Bare-handed operation serves as a reference for human performance, although it does not directly produce robot-executable demonstrations.

One experienced operator completed five 5-min windows per task and condition. Shown in fig. 7, WUDI achieved 7.50, 3.62, and 7.71 times the mean completions of vision-based teleoperation on Zipper, Coin, and Lid, respectively. Two-sided exact permutation tests on window-level counts, with Holm correction across the three tasks, yielded adjusted $p = 0.0238$ for each comparison. These results support higher throughput for this operator and setup.

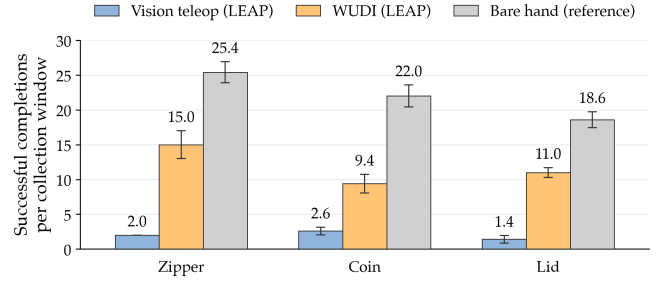


Fig. 7: Mean number of successful completions per 5-min collection window (five windows per condition); error bars indicate sample standard deviation across windows.

F. Design Extensions to Additional New Hands



Fig. 8: Five WUDI passive-hand and linkage designs (CAD and sim-verified, not fabricated) for different hands.

Beyond WUJI Hand v1 and LEAP Hand, we design five additional passive-hand and linkage designs (Fig. 8), each requiring only embodiment-specific coupler dimensions and mount adjustments. These designs are assembled in MuJoCo and checked for linkage closure but not fabricated because we lack the corresponding robot hands. We will release all seven designs as open-source hardware.

VI. LIMITATIONS AND OUTLOOK

Like many other dexterous data collection systems, WUDI is not without limitations: First, differences between target-hand and exoskeleton geometry can require fixing thumb joints, reducing articulation and workspace. The evaluated LEAP configuration supports the tested tasks but does not establish broader in-hand dexterity. Thumb mounts and coupler dimensions also require manual design. Second, the exoskeleton and linkages restrict access to confined spaces. Our collection setup avoids visual adaptation using wrist-mounted observations, so third-person views may require visual inpainting. Third, the rigid interface limits fit, particularly for larger hands, and requires donning adjustments. Concentrated weight and an elevated arm posture can cause discomfort during prolonged use. No participant succeeded on Squeeze or Unbuckle; we retain these tasks in OTHER because bare-hand complete them readily (Fig. 1); they expose the current gap between interface-hand and human dexterity. Unlike interfaces designed around a particular hand [1]–[3], WUDI must accommodate substantially different kinematic structures and geometries. We therefore prioritize cross-hand applicability over maximizing fit or articulation for any single platform. This generality may come at the cost of lower data-collection efficiency or joint-reconstruction accuracy, but quantifying these trade-offs requires direct evaluation on the corresponding dexterous-hand platforms.

Future work will improve fit, weight distribution, durability, and thumb-linkage design, and quantify motion coverage and adaptation effort. We also plan to integrate tactile sensing and haptic feedback to record contact forces, addressing the grip-force failures observed on LEAP-Lid.

Overall, WUDi demonstrates that a shared sensing interface can support demonstration collection across robot hands through interchangeable passive components. Our user study and policy experiments establish functional capability within the supported workspace, while OTHER provides a common benchmark for evaluating human-operated manipulation and autonomous policies across embodiments.

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